INTRODUCTION TO R



PRESTIGE WINE BUTLER

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AGENDA

FIRST INTRODUCTION TO THE DATASET

01

Introduction to the dataset
Sources, context, size, variables, etc., tables and graphs

02

Why choose this dataset ?
Purpose, storytelling

03

First look of the application
Visual of the wanted result

WINE PRESTIGE BUTLER

We are delivering a prestige service In teh finding of excellence wine

01

#We will begin this report, by introducing our dataset with basic facts.
First, let's talk about the source, context, identity of the database.

- **Source** : The database was found on kaggle.com. Kaggle was the first site on which I have decided to look for a database. This choice is the result of the recommendations made in class and after speaking to a few students.
- **Context** : The aim of finding a database is to later on be able to process it, to transform it into something that will later on be used within an application in order for the user to have the most precise result.

But more importantly, after having a father passionate about wine and after growing up in a famous wine region : Bourgogne, I wondered how could I create an application that identify for the users that are not wine experts. Which wine is perfect for their tastes, budget, preferences, etc. ? The objective of this dataset is to offer the result you could have from a five stars sommelier.

The dataset that we will name : prestige_wine_butler or pwb contains:

- 17 000 lines
- 9 variables composed of 6 categorical , 1 "normal text" variable, 1 ordinal, 1 interval with natural numbers variables.

Below is stated the title of each variable, the nature of the variable and the composition of each categories within the variables.

Name of the variable	Type of the variable	Nb of categories within the variable
Continent	Categorical Level_1	4
Country	Categorical Level_2	8
Name of the domain	Normal text	17003
Rate	Ordinal	10
Price	Interval with natural numbers	268
Province	Categorical Level_3	43
Region	Categorical Level_4	601
Variety of grapes	Categorical Level_5	121
Color of grapes	Categorical Level_6	5

The cleaning part of the database was a very important step of the process of working with the database.
The reason for this action is the result of wanting to have a qualitative findings for the user and therefore to avoid having results incomplete.

The methodology used was the one following:

1. **Step 1:** cleaning of every row with incomplete columns
2. **Step 2:** deleting all the numerical values in the column "Name of the domain"
   a. *cf. Some rows had as a name "1". If that was the case, I deleted the whole row.*
3. **Step 3:** deleting all the special characters or wrongly written names
   a. *cf. Some rows had content with "@". If that was the case, I deleted the whole row.*
4. **Step 4:** deleting all the wine lines that had a score rate lower than 90. Indeed the aim of this dataset is not the large extent of the possible findings but this is all about qualitative findings. Meaning we have only kept the wine with the highest score possible (from 90 to 100). Prestige Wine Butler is about excellence finding strategy.
5. **Step 5:** Transforming all the values of the price in numerical as it was read as text before.
6. **Step 6:** Adding 2 categorical columns : Continent and color of the grapes in order to give to the user the best user-experience possible as some people might not have a clear idea of the type of wine they want. Therefore with the possibility to select general criterias such as continent or color of the grapes will be useful.

After this cleaning process we have gone from having 150k lines to 17k.

You will find below the final result of the database.

Continent	Country	Name of the domain	Rate	Price	Province	Region	Variety of grapes	Color of the grapes
North America	US	Martha's Vineyard	96	€235	California	Napa Valley	Cabernet Sauvignon	Red
Europe	Spain	Carodorum Selección Especial Reserva	96	€110	Northern Spain	Toro	Tinta de Toro	Red
North America	US	Special Selected Late Harvest	96	€90	California	Knights Valley	Sauvignon Blanc	White
North America	US	Reserve	96	€65	Oregon	Willamette Valley	Pinot Noir	Red
Europe	France	La Brûlade	95	€66	Provence	Bandol	Provence red blend	Red
Europe	Spain	Numanthia	95	€73	Northern Spain	Toro	Tinta de Toro	Red
Europe	Spain	San Román	95	€65	Northern Spain	Toro	Tinta de Toro	Red
Europe	Spain	Carodorum Único Crianza	95	€110	Northern Spain	Toro	Tinta de Toro	Red

IDENTITY OF THE DATASET

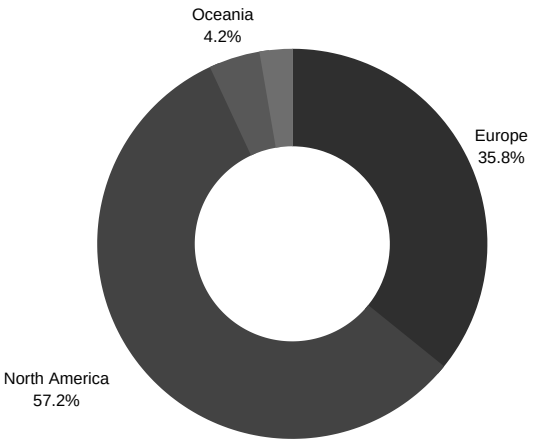


FIG. COMPOSITION OF THE CONTINENT VARIABLE (TOP 3 SHOWN)

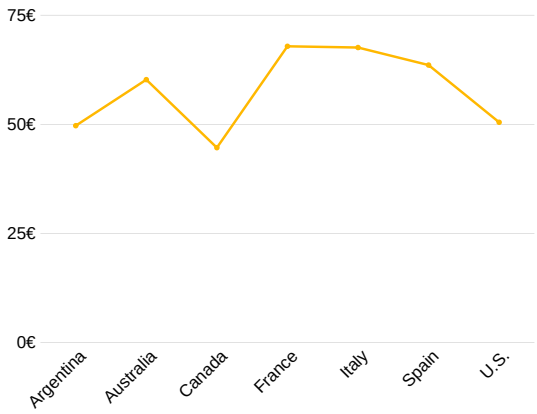


FIG. AVERAGE PRICE PER COUNTRY

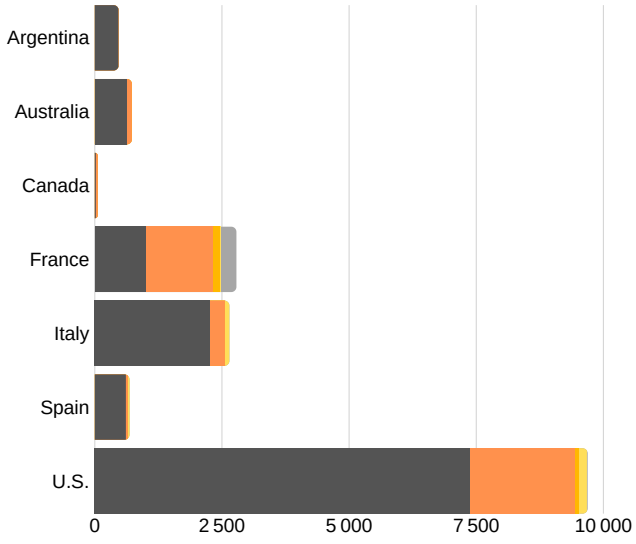


FIG. COMPOSITION OF COLOR OF GRAPES PER COUNTRY



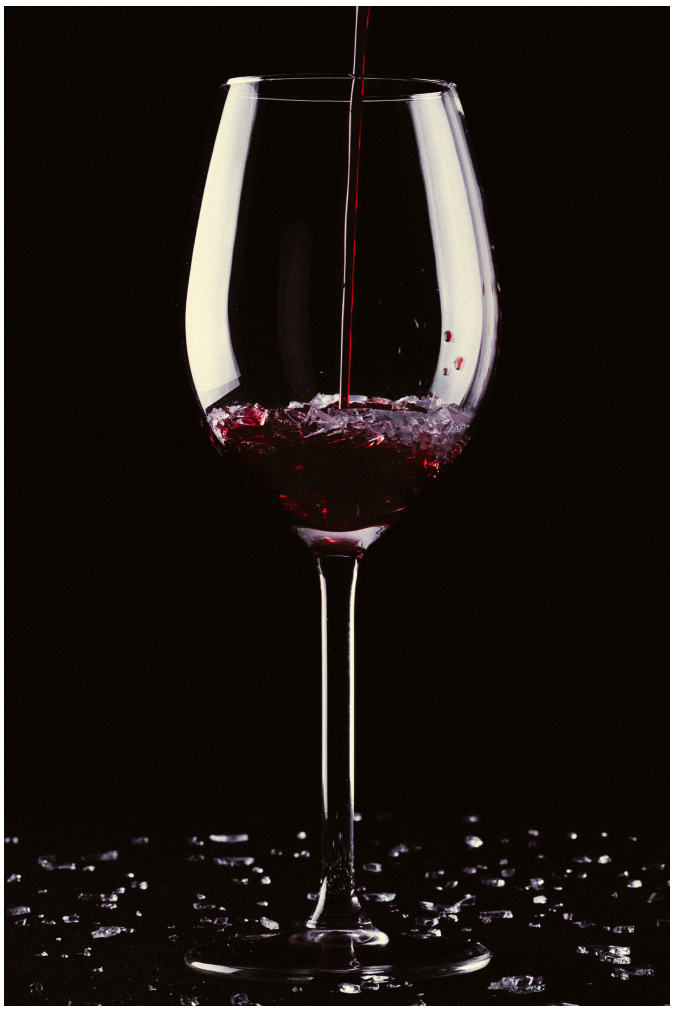
Regarding the actual content of pwb, here are some simple statistics to be found of the left part of the page.

There is a clear dominance of the American wine with more than 50% of the general composition of the database.

There is also a strong presence of red wine throughout the different region.

Generally speaking, the lowest prices for wine are mostly from Canada.

Thanks to all these variables, the user will be able to find the most personalized findings.



02 STORY TELLING, PURPOSE

Like said before, growing up in a wine region has made me more sensitive to the quality/history of the wine in general. The general idea behind this database is to be able to deliver the best wine results from a qualitative dataset with specific criterias in order for the user to get the service she/he deserves.

The data is originally from WineEnthousiast (June, 2017) and has been transformed by the data scientist Hitesh Palamada.

What is our aim ?

- To offer an experience to our users: simulate an interaction between a customer and a sommelier in order for the client to have the best choice given in accordance to his/her preferences.

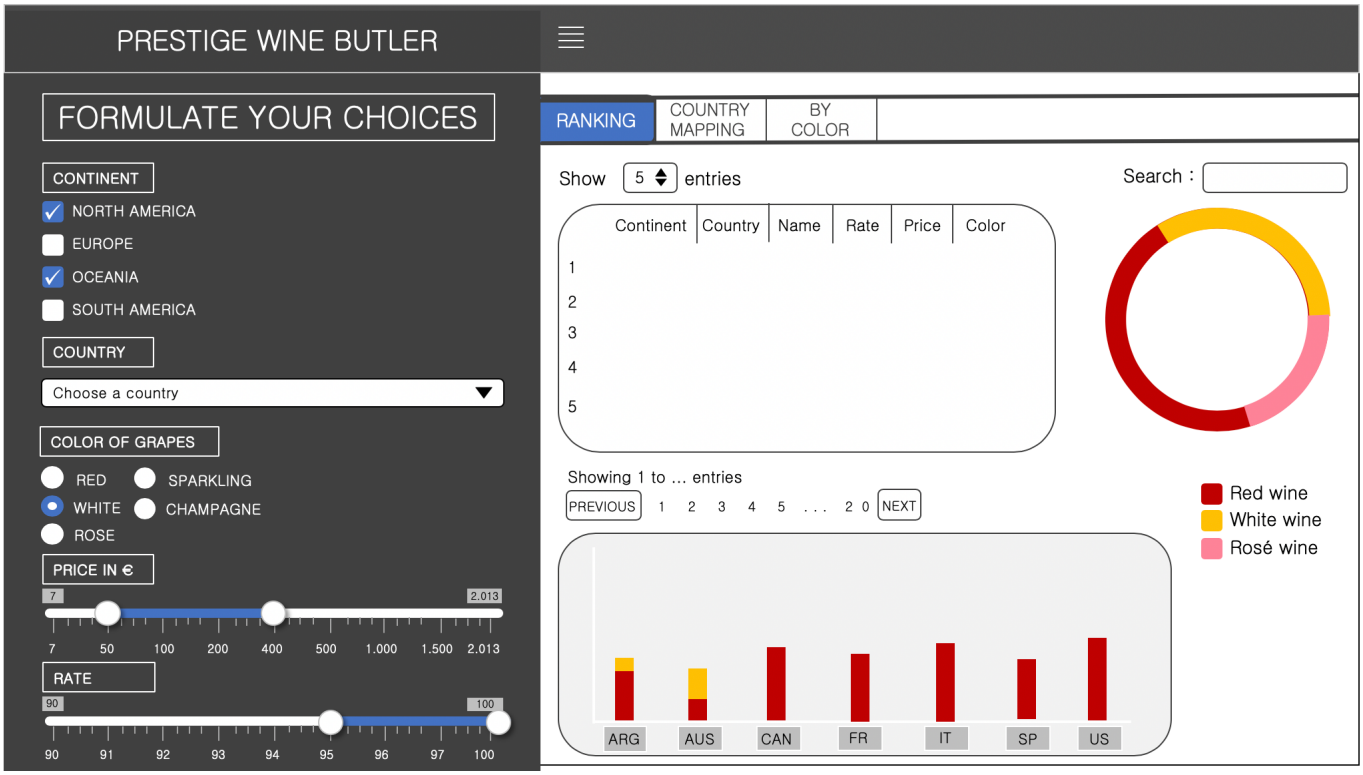
The idea is to be able to show the best wine someone can find based on their different criterias (continent, region, color of the grapes, etc.).

This is why we will offer different kind of services for the user in order to have the most personalized experience:

- The user will be able to select from different options : continent, price, rate, color, variety of grapes, etc. and will get the top 5 results shown.
- The user will be able to select from different options : price, rate, color, variety, etc. to be able to see on a interactive map where are the wine that he/she wants to find.
- The user will have access to predetermined graphs and on the top will be presented on different blocs the number of findings of the best country :
  - Best wine producing countries - rate point wise
  - Most expensive wine / Least expensive wine



03 VISUAL OF THE RESULT



Above is a first version of what the application could look like. Keeping in mind that some modifications will be made for the final version.

The idea is to have the left part, everything that is widget related, this is the "play area" for the user. From the top to the bottom, the logic is to be more and more precise in the research.

The first tab has as main focus the ranking which will appear in different way :
- First with a classic table ranking showing the top 5 results
- Secondly, with a column graphic which will provide a visual aspect for the user the repartition of the top 5 (or more) results per country and per color.
- Thirdly, we will show with a circle graphic, the repartition of the color for the findings.

The second tab, will be an interactive map In which the user will be able to see on a map where are the best wine that he/she has chosen (meaning the results of his/her query). By Interactive, we mean that If the mouth goes through the touching point, the name of the wine will appear.

The third tab, will be a very simplistic tab in which the user will find 3 blocs which will synthesize the volume found per color of the wine with an icon next to it.